



1

3

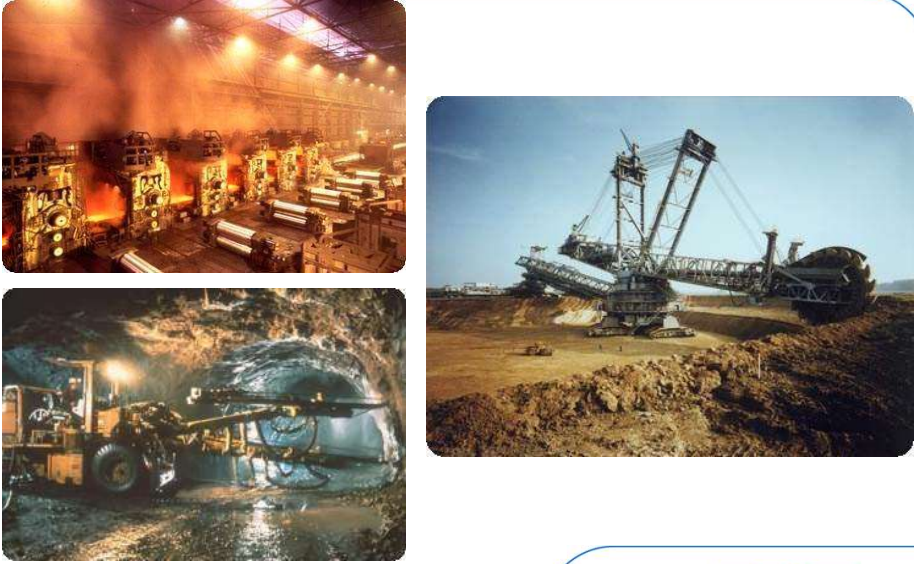
Contamination and cleanliness

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2

Contamination/cleanliness



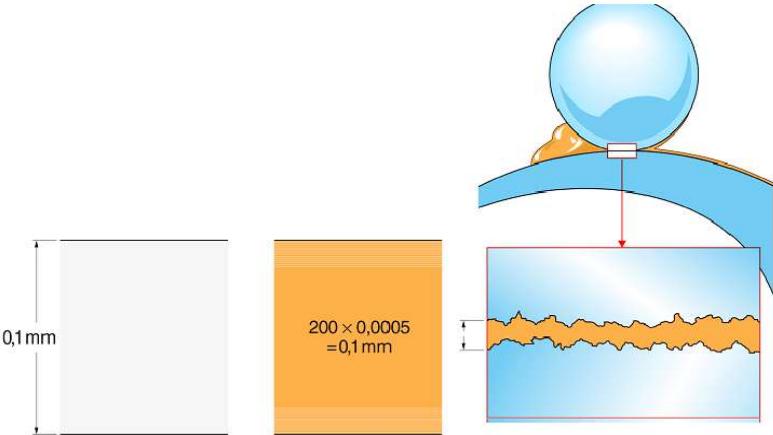
The top-left image shows a smelter or refinery with bright orange molten metal. The bottom-left image shows a tunnel boring machine (TBM) cutting through rock. The right image shows a large excavator working on a construction site.

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Contamination...



The diagram shows a cross-section of a bearing. On the left, a grey square represents a contamination layer with a height of 0,1mm. In the middle, an orange square represents a contamination layer with a height of $200 \times 0,0005 = 0,1\text{mm}$. On the right, a blue circle represents a bearing ball, and a red arrow points to a cross-section of the bearing showing the contamination layer.

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Contamination...

The diagram on the left illustrates the process of contamination in a bearing. It shows a large particle (0.005 mm) being introduced into a bearing with a surface finish of 0.0005 mm. Below this, four sequential cross-sectional views show the particle moving through the oil film, causing surface damage and debris. To the right, four micrographs show the progression of damage: from a small initial mark to a large, deep crater.

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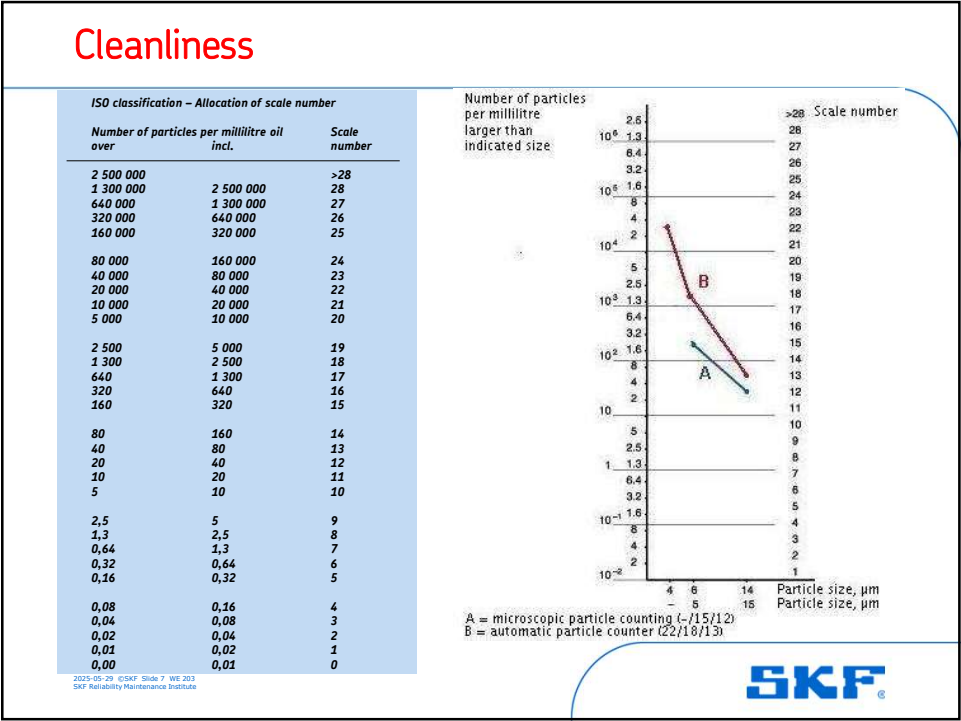
Contamination

This diagram compares the sizes of various contaminants relative to a bearing's oil film. The oil film is shown as a thin layer between two surfaces. The contaminants are represented by different shapes and sizes:

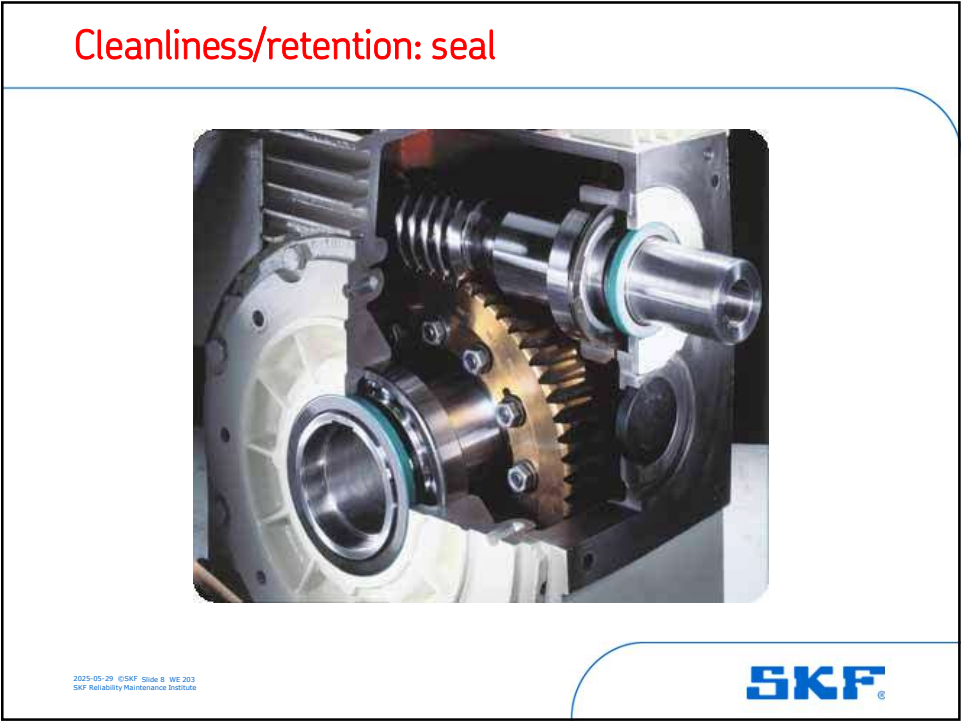
- TOBACCO SMOKE: 100 MICRO INCH (2,5 μm)
- INDUSTRIAL SMOKE PARTICLE: 250 MICRO INCH (6,5 μm)
- DUST PARTICLE: 1000 MICRO INCH (25 μm)
- HUMAN HAIR: 3000 MICRO INCH (76 μm)
- FINGER PRINT: 500 MICRO INCH (13 μm)
- OIL FILM: 15 MICRO INCH (0,4 μm)

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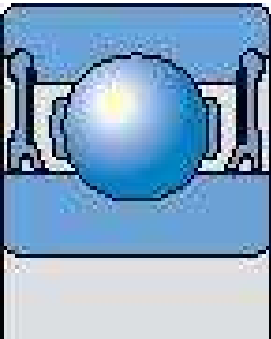
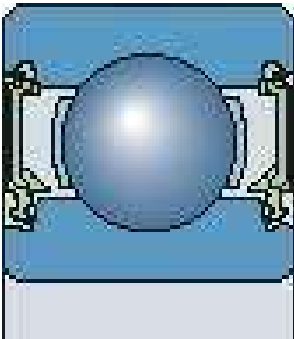
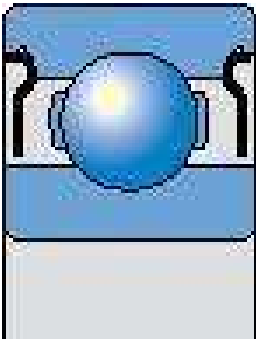


7



8

Seals/shields: integral...



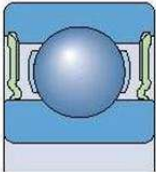
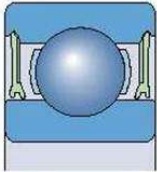
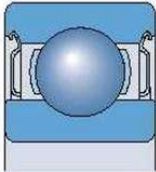
ShieldsContact SealsLow-Friction Seal

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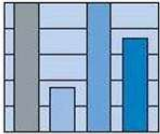
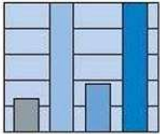
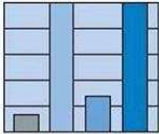
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Seals/shields: integral



-Z and -2Z-RZ and -2RZ-RS1 and -2RS1

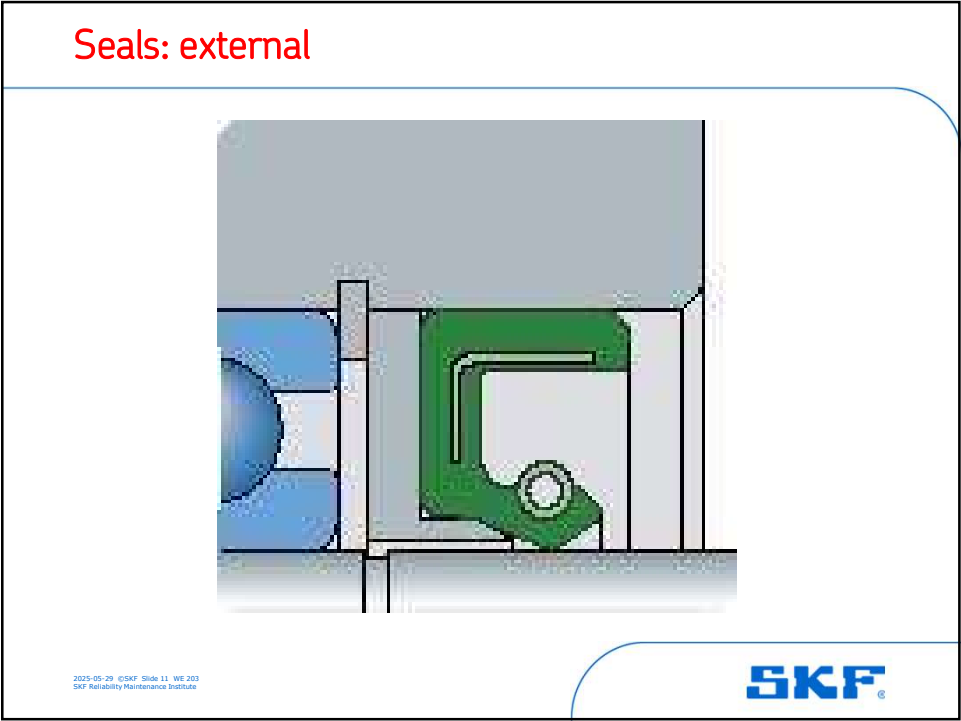


 Contaminant exclusion	 Silent running	 Grease retention	 Speed capability
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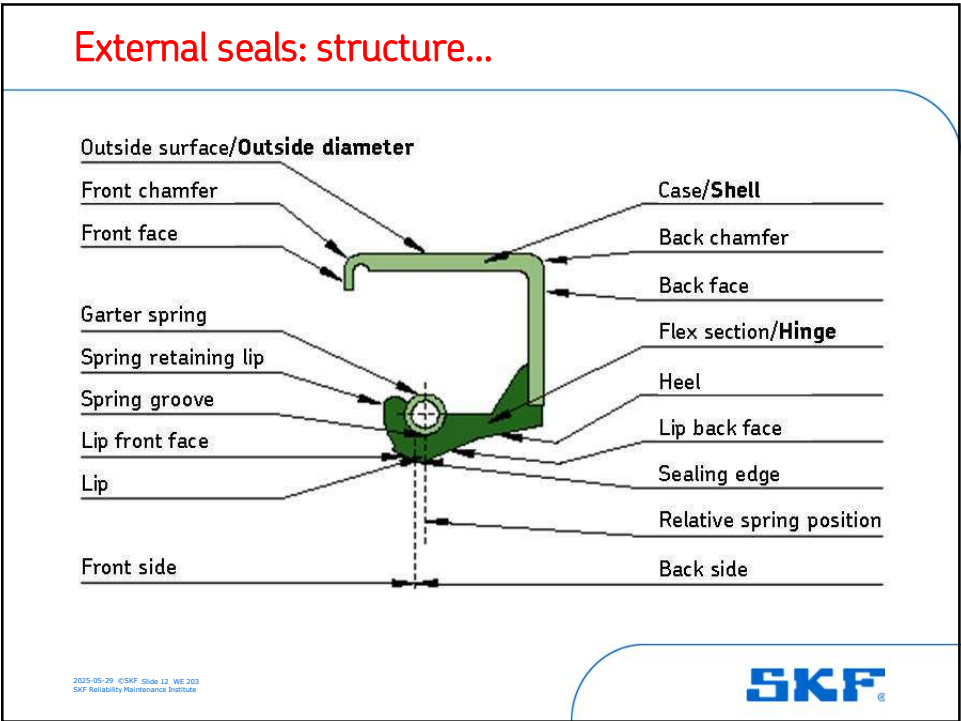
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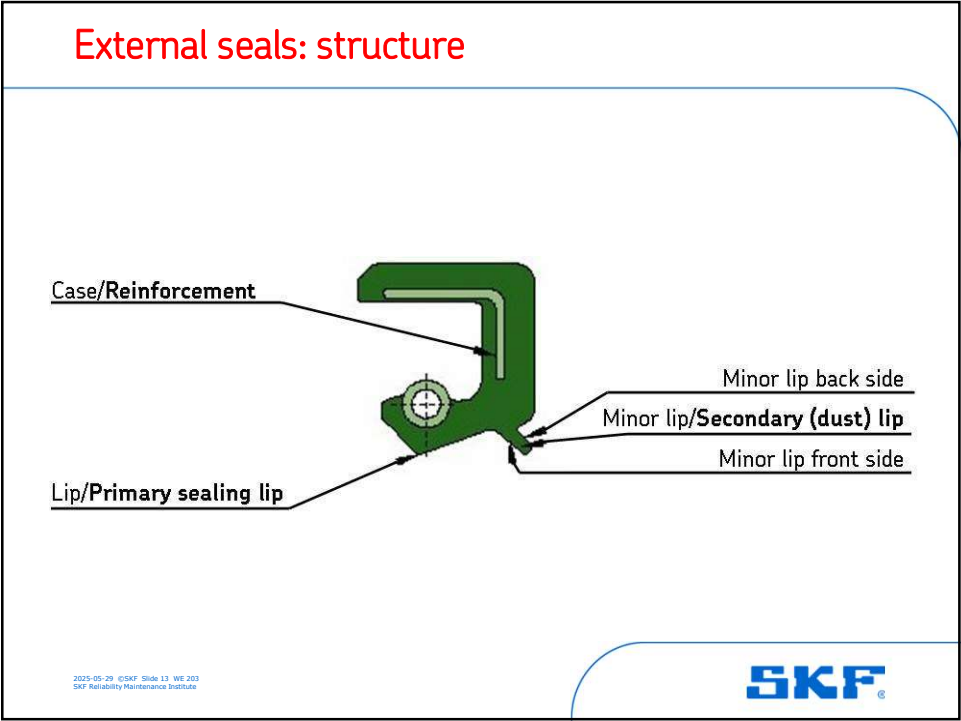
10



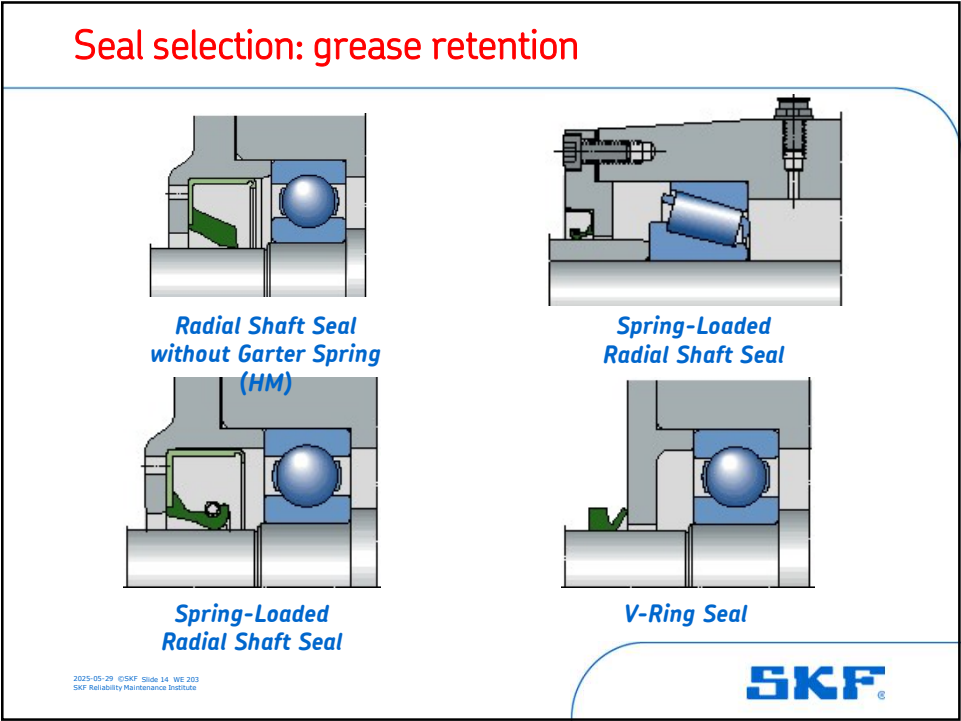
11



12

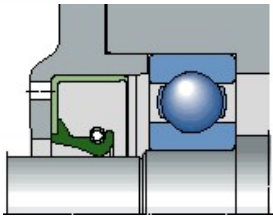


13

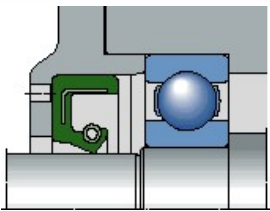


14

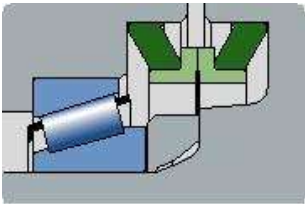
Seal selection: oil retention



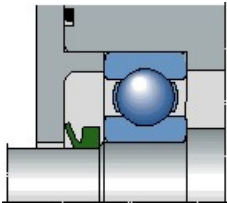
CRW1



HMS4



HDDF Mechanical Seal



V-Ring Seal

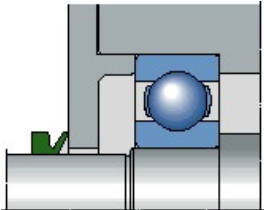


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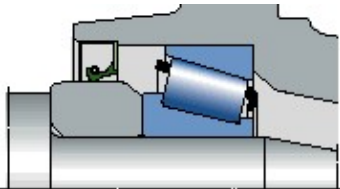


15

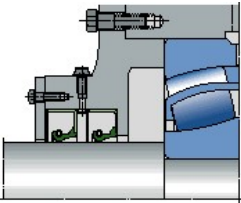
Seal selection: contaminants exclusion



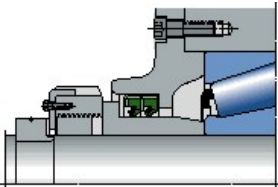
1) V Ring



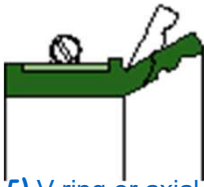
2) CRW1 / CRWH1



3) Seal arrange in tandem



4) Double lips in tandem



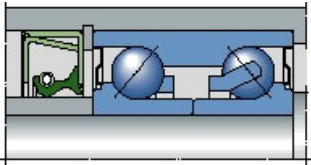
5) V-ring or axial clamp seal

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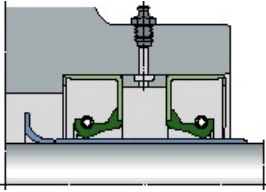
Seal selection: exclusion and retention



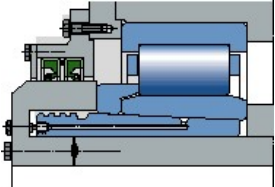
CRWHA1 Seal



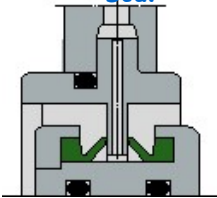
HMSA7 Seal



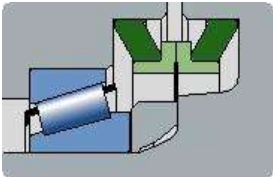
Two CRW1 Seals



Two HDS3 Seals



Two Opposing V-Ring Seals



HDDF Mechanical Seal

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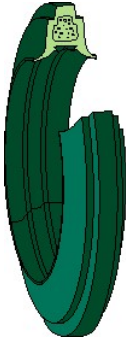
17

Seal selection: standard housings...

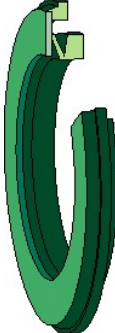




TSN .. L
four-lip seals



TSN .. G
double-lip seals



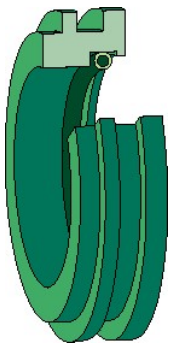
TSN .. A
V-ring seals

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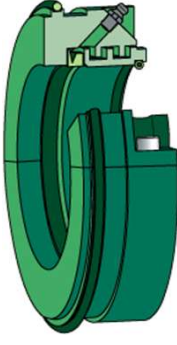


18

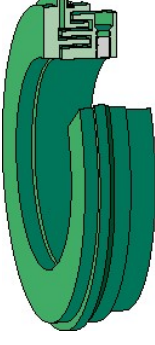
Seal selection: standard housings



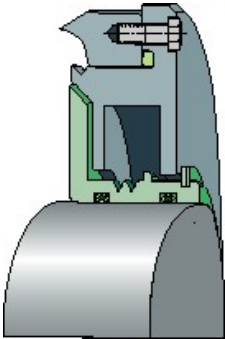
TSN...S
Labyrinth seal



TSN..ND
Taconite heavy-duty seal



TSN..NC
Taconite heavy-duty seal



TSN..UL
Seals for oil lubrication

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Module closeout

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Name: _____ <Optional to Include Your Name>

WE 203 – Lubrication in Rolling Element Bearings
XX/XX/2007

Module Number & Title: 3. Contamination and Cleanliness

My most important question on this module is:
.....
.....
.....

I did not understand the following:
.....
.....
.....

I do not agree with the following:
.....
.....
.....

My top three learnings from this module (referenced to your expectations) are:

1.
.....
.....

2.
.....
.....

3.
.....
.....

Is this learning related to
your own expectations?

Yes	No
Yes	No
Yes	No

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The SKF logo is displayed in a large, bold, blue font. The letters are stylized, with the 'S' and 'K' being particularly prominent. A small registered trademark symbol (®) is located to the right of the 'F'. The logo is centered within a white rectangular area that has rounded corners and is framed by a thin black border. There are some faint red lines and shapes around the logo, possibly indicating a design or a scanning artifact.

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Practical Application in Bearing
Lubrication Course

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